Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	03 October 2026	**Team ID**	SWTID-2026-3942
Project Name	Import Data using Transform Maps (Spreadsheet)	**Maximum Marks**	4 Marks

Technical Architecture:

Spreadsheet (.xlsx) → Employee Import (Import Set table u__employee_import) → Sample Spreadsheet Import (Transform Map) → Employee Test (u_employee_test) → Reports → Employee Analytics Dashboards

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	ServiceNow web interface used to create tables, load the spreadsheet, configure the Transform Map, and build reports and the dashboard.	ServiceNow Web UI (Next Experience)
2	Application Logic-1	Loads the structured employee spreadsheet into the staging Import Set table.	ServiceNow Import Sets / Load Data
3	Application Logic-2	Maps source fields from Employee Import to Employee Test and runs the transformation.	ServiceNow Transform Maps
4	Application Logic-3	Matches existing employees on Employee ID so records are updated rather than duplicated; field scripts convert values when needed.	Coalesce, server-side JavaScript (Glide API)
5	Database	Stores the employee records in the custom target table with Employee ID, Employee Name, Email, Department and Location.	ServiceNow table: Employee Test (u_employee_test)
6	Cloud Database	Application tables live in the ServiceNow platform's own relational database; no separate cloud database is used.	ServiceNow platform database
7	File Storage	The source spreadsheet is prepared in Google Sheets, downloaded as .xlsx, and attached during Load Data.	Google Sheets / Excel (.xlsx)
8	External API-1	Optional: the platform's Import Set REST API can insert a staged record and run the Transform Map.	ServiceNow REST API (/api/now/import)
9	External API-2	Optional: the Table API can list target-table records for verification.	ServiceNow REST API (/api/now/table)
10	Machine Learning Model	No machine learning model is used; the Transform Map is rule-based.	Not applicable
11	Infrastructure (Server / Cloud)	The whole import workflow runs on a ServiceNow Personal Developer Instance (PDI).	ServiceNow Cloud (PDI)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	None; the solution is configured on the ServiceNow platform without external frameworks.	Not applicable
2	Security	Username and password sign-in with role-based access (admin, import_admin, import_transformer); ACLs protect the import and target tables; dashboard sharing can be limited by users, groups and roles.	ServiceNow Roles / ACLs / Users / Groups
3	Scalable Architecture	Bulk employee data goes through Import Sets and a Transform Map, so larger or repeated imports use the same configuration.	ServiceNow Import Sets & Transform Maps
4	Availability	Reports and the dashboard give a central view of employee records inside the platform, available to permitted users.	ServiceNow Reports & Platform Analytics
5	Performance	One bulk import replaces manual entry; the transform processes all staged rows in a single run.	Import Sets & Transform Maps

Source basis: Entries are taken from this team’s project documentation (FSD) and ServiceNow configuration.